

EXECUTIVE TQM GURUS & STRATEGIC QUALITY FRAMEWORKS

Comprehensive Industry Reference Guide: Deming, Shewhart, Juran, Crosby, Feigenbaum, Ishikawa, Taguchi & Hoshin Kanri

A. STATISTICAL QUALITY CONTROL PIONEERS (SHEWHART & DEMING)

TQM GURU	HISTORICAL TITLE & DISTINCTION	CORE QUALITY PHILOSOPHY	KEY TOOLS & FRAMEWORKS	MANUFACTURING IMPACT
Walter A. Shewhart	Father of Statistical Quality Control (SQC)	- Statistical process variation management - Common Causes vs. Special Causes of variation - Foundational principles of Process Control	- Control Charts ($\bar{X}$-R Charts, p-charts) 3σ Natural Process Limits - Specification vs. Control Boundaries	Eliminated guesswork in shop-floor machining tolerances using empirical statistical limits.
W. Edwards Deming	Father of Modern Quality Management	14 Points for Top Management Transformation - System of Profound Knowledge (SoPK) - Shift from inspection to system design - Elimination of numerical quotas & fear	- PDCA / PDSA Cycle (Plan-Do-Check/Study-Act) - Red Bead Experiment - Funnel Experiment	Spearheaded Japan's post-WWII industrial renaissance (Foundation of Toyota Production System).

B. STRATEGIC & FINANCIAL QUALITY LEADERSHIP (JURAN & CROSBY)

TQM GURU	HISTORICAL TITLE & DISTINCTION	CORE QUALITY PHILOSOPHY	KEY TOOLS & FRAMEWORKS	MANUFACTURING IMPACT
Joseph M. Juran	Pioneer of Quality Management & Cost of Quality	- Quality defined as "Fitness for Use" - Cost of Quality (COQ) Framework - Pareto Principle applied to Quality ("Vital few vs. Useful many")	- Juran's Quality Trilogy: Quality Planning, Quality Control, Quality Improvement - COQ Breakdown: Prevention, Appraisal, Internal & External Failures	Linked defect metrics directly to board-level financial performance and operational P&L.
Philip B. Crosby	Advocate of Zero Defects & Price of Non-Conformance	"Zero Defects" Mindset & Philosophy "Quality is Free" (Prevention saves more than it costs) - Quality is Conformance to Requirements	4 Absolutes of Quality Management - Price of Non-Conformance (PONC) vs. Price of Conformance (POC) 14-Step Quality Improvement Program	Enforced zero-defect standards across aerospace and high-precision defense manufacturing.

C. TOTAL QUALITY CONTROL, ROOT CAUSE & ROBUST DESIGN (FEIGENBAUM, ISHIKAWA, TAGUCHI)

TQM GURU	HISTORICAL TITLE & DISTINCTION	CORE QUALITY PHILOSOPHY	KEY TOOLS & FRAMEWORKS	MANUFACTURING IMPACT
Armand V. Feigenbaum	Father of Total Quality Control (TQC) & TQM Originator	- Total Quality Control (TQC) as company-wide accountability "Hidden Factory" Concept (capacity wasted on rework) - Customer-defined total quality system	- Total Quality System Architecture - Cross-Departmental Quality Ownership Matrix (R&D to Purchase/Sales)	Structured the modern enterprise-wide Total Quality Management (TQM) architecture.
Kaoru Ishikawa	Pioneer of Shop-Floor Quality & CWQC	- Democratization of quality control to shop-floor operators - Company-Wide Quality Control (CWQC) - Quality Circles (QC Circles)	- Cause & Effect Diagram (Fishbone / Ishikawa) 7 Basic QC Tools (Check Sheet, Pareto, Fishbone, Histogram, Scatter Plot, Control Chart, Stratification)	Empowered front-line workers to independently solve shop-floor defects through Kaizen.
Genichi Taguchi	Pioneer of Robust Design & Loss Function Mechanics	- Quality Loss Function: Financial loss occurs upon any deviation from nominal target - Robust Engineering & Off-Line Quality Control	- Taguchi Loss Function: $L(y) = k(y - m)^2$ - Design of Experiments (DOE) Orthogonal Arrays - Signal-to-Noise (S/N) Ratio	Minimized production variation during product design stage prior to mass manufacturing.

D. STRATEGIC ALIGNMENT & POLICY DEPLOYMENT (HOSHIN KANRI & X-MATRIX)

FRAMEWORK / TOOL	ORIGIN & PURPOSE	CORE PRINCIPLES & STRATEGIC ALIGNMENT	IMPLEMENTATION MECHANICS	MANUFACTURING IMPACT
Hoshin Kanri (Policy Deployment)	Yoji Akao & Shigeru Mizuno (Japan, 1960s)	- Alignment of C-suite vision down to shop-floor daily execution "Catchball" Process (two-way multi-level negotiation) - Breakthrough targets vs. routine daily management	7-Step Hoshin Planning Cycle - Catchball Feedback Loops - Annual & 3-to-5 Year Strategic Milestones	Eliminates departmental silos and aligns plant-level Kaizen directly with C-suite targets.

Framework / Tool	Origin & Purpose	Core Principles & Strategic Alignment	Implementation Mechanics	Manufacturing Impact
Hoshin Kanri X-Matrix	Visual Strategic Execution Tool for TQM	- Single-page visual alignment linking 4 key quadrants: - Long-Term Strategies (3-5 Yrs) - Annual Objectives (1 Yr) - Top Improvement Projects - Operational KPIs & Metrics	4-Quadrant Intersecting Matrix - Correlation Mapping (Strong / Secondary Dots) - Primary & Secondary Ownership Assignment	Provides line-of-sight visibility for plant managers and ISO/IATF audit executives.

EXECUTIVE TQM FRAMEWORK MATRIX • SYNTHESIS SUMMARY

1. STATISTICAL QUALITY - Shewhart: 3σ Limits & Control Charts - Deming: PDCA & Profound Knowledge	2. FINANCIAL QUALITY - Juran: Trilogy & Cost of Quality (COQ) - Crosby: Zero Defects & PONC	3. ORGANIZATIONAL TQM - Feigenbaum: TQC & Hidden Factory - Ishikawa: 7 QC Tools & Circles	4. STRATEGIC DEPLOYMENT - Taguchi: $L(y) = k(y - m)^2$ - Hoshin Kanri: X-Matrix Alignment
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